

EE6406

Analytic and Ensemble Machine Learning (Lecture 2: Data Preprocessing)

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EE6406: Analytic and Ensemble Machine Learning

Lecture 2

Data Preprocessing

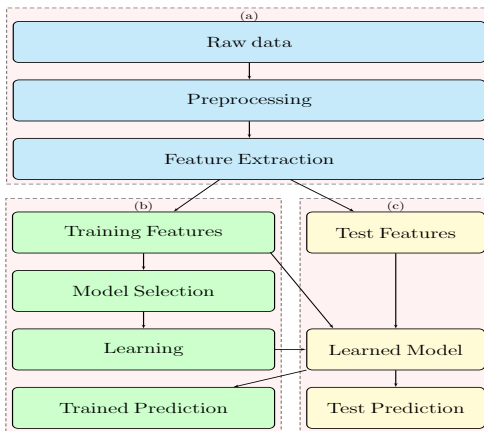
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 - Nominal, Ordinal and Interval Data
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Lecture 2

Learning Pipeline

Pattern Recognition Pipeline



Data Preprocessing (Focus of This Lecture)

- Prepares the data for meaningful representation.
- **Normalization**: the process of removing non-representative or redundant parts of data and make relevant adjustment. The main purpose of data normalization is to prevent anomalies from dominating the analysis.
- **Data conversion**: conversion among different data types such as *nominal*, *ordinal*, *interval*, and *ratio data* attributes.

Feature Extraction

- *Feature extraction* is the process of extracting informative, relevant and non-redundant components from the bulk of data to facilitate subsequent learning.
- Essentially, it involves a **reduction of the amount of resources required** for describing a large set of data.
- This process often largely reduces the dimension of data and is thus related to **dimension reduction**.
 - **feature extraction**: often involves the process of data transformation.
 - **feature selection**: also known as variable selection, is the process of selecting a subset of relevant features (variables, predictors) for use in model construction.

Feature Extraction

Approaches to feature extraction can be broadly categorized into the approach of generic *dimension reduction* and the approach of extracting *heuristic* or *semantic features*.

- **Dimension reduction:** *principal component analysis (PCA), independent component analysis (ICA), isomap, multilinear subspace learning and autoencoder.*
- **Extracting semantic features:** *edge detection, corner detection, blob detection, ridge detection and scale-invariant feature transform.*

Learning

Essentially, machine learning in our context can be summarized as learning a function f that maps input variables to output variables i.e.,

$$Y = f(X)$$

where X and Y are respectively generic symbols for input and output variables which can take either scalar or vector quantities.

The form of f , if it exists, is **unknown in practice** and the job is to evaluate those algorithms at hand that best describe the problem.

Remark: Symbols on this slide are generic (abstract variables) and do not denote specific scalar, vector, or matrix data types.

Decision and Performance Evaluation

Prediction and Category Decision

- After the training phase, the optimized model f is applied to unseen data X_t to generate the predicted output $Y_t = f(X_t)$.
- For classification, a threshold decision is applied to the output $Y_t = f(X_t)$ to determine the object category.

Remark: Symbols on this slide are generic (abstract variables) and do not denote specific scalar, vector, or matrix data types.

Decision and Performance Evaluation

Performance Evaluation

- In practical applications, the volume of training data is frequently constrained by acquisition challenges and budgetary limits.
- The model's performance on unseen data thus depends on its predictivity or *generalization* capability when trained on limited data samples.

Decision and Performance Evaluation

Performance Evaluation

- Although the underlying data distribution is unknown, model performance can still be assessed through an **appropriate evaluation framework**. This is typically achieved by **withholding a portion of the dataset** to **evaluate** the generalization capability of the learning algorithm.
 - To maximize the utility of all available data, an **N -fold cross-validation** scheme can be employed. The entire dataset is partitioned into N equally sized folds; $N - 1$ folds are used for training, while the remaining fold is reserved for testing. This process is repeated iteratively so that every fold serves as the test set exactly once, with the final performance reported as the average across all N trials.

Lecture 2

What is Data?

What is Data?

According to <https://en.wikipedia.org/wiki/Data>:

“In common usage, **data** are a collection of **discrete or continuous values** that **convey information**, describing the quantity, quality, fact, statistics, other basic units of meaning, or simply sequences of symbols that may be further interpreted formally. A datum is an individual value in a collection of data. ”

Example 2.1

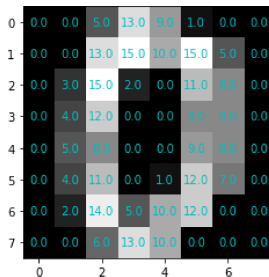


Figure 2: Left panel: samples of Optdigit dataset, right panel: the values in digit-0 sample.

Example 2.2

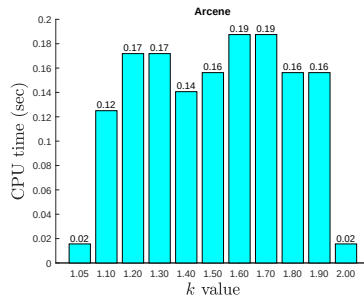
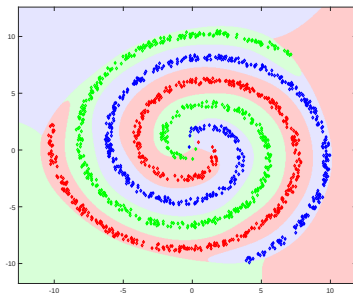
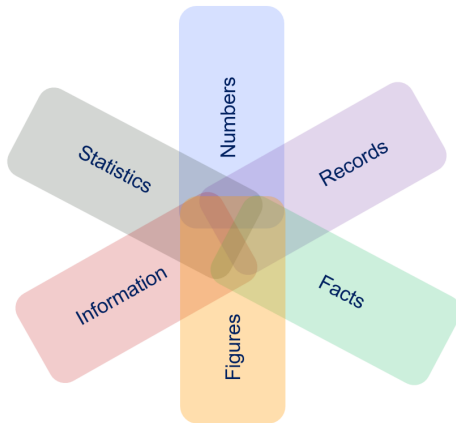


Figure 3: Left panel: a spiral pattern, right panel: a bargraph

What is data?

- **Continuous, Discrete**
- **Ordinal**
- **Categorical**
- **Missing**
- **Censored**



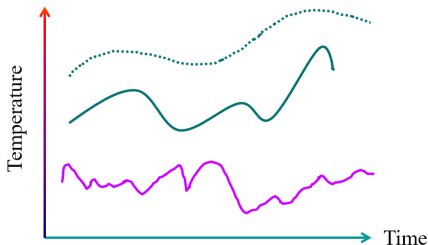
Lecture 2

Data Types

Example 2.3

Continuous, discrete variables

- Continuous variables are anything measured on a quantitative scale that could be any **fractional number**.
 - An example would be something like weight measured in kg.
- Discrete variables are numeric variables that have a **countable number** of values between any two values.



Example 2.4

Ordinal data

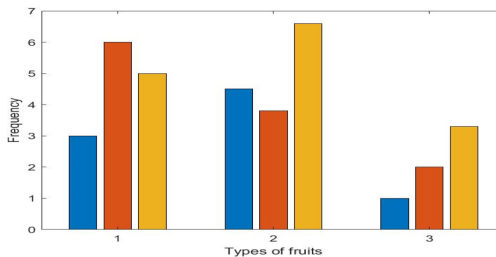
- Ordinal data are data that have a fixed, small number (< 100) of possible values, called levels, that are **ordered** (or ranked).
- Example: survey responses

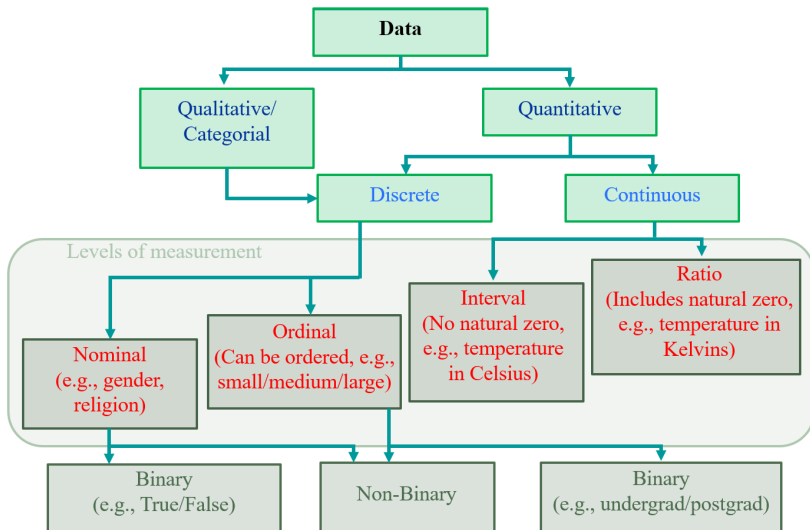
Excellent	Good	Average	Fair	Poor
☒	☐	☐	☐	☐

Example 2.5

Categorical data

- Categorical data are data where there are multiple categories, but they **are not ordered**.
- Examples: gender; blood type, name of fruits and their production regions etc.





Data Types

- **Nominal data:** Nominal data attributes are also called categorical or qualitative data attributes. With some exceptions, the categories are distinct but have **no explicit ranking or order**. Such attributes often appear in natural settings as the attribute label or name form.
- **Ordinal data:** For data of ordinal scale, the exact numerical value has no significance over the ranking order that it carries. In other words, the categories **have an explicit order**, but **the difference between them is not necessarily uniform or measurable**. Such data is often seen in social and medical sciences when certain survey or health conditions are coded in importance level or strength level.
- **Interval and ratio data:** Both interval attributes and ratio attributes of data are represented by numerics. However, they are different in terms of whether the natural (absolute) zero-point is included. The ratio scale has an absolute zero-point. For example, age, height and weight never fall below zero. For the interval scale, the zero-point is arbitrary where negative values is permitted.

Nominal data

Examples

- **With no intrinsic order:** gender (male or female), race, blood type, place, identification number, name of fishes, name of fruits, and etc. These data can be represented in summary form, either in the form of percentage or in terms of frequency of occurrence. However no *mean* value regarding the labels can be defined.
- **With intrinsic order:** color where the underlying electromagnetic wavelength can be measured, age where the measurement is time based, and educational level where an intrinsic order can be established.

Nominal data

Numeric Conversion / Data encoding

- **Arbitrary assignment:** Assign arbitrary numbers to represent these attributes.
 - For example, one can assign a '1' value for male and a '2' value for female for the case of gender attribute. However, the label which is assigned a **higher value may have a greater influence** than the one with a lower value when extremely large values and extremely small values are involved along the computational process.
 - Moreover, the **intrinsic relationship** among the nominal attributes may be lost when the numerical assignment is arbitrary. For example, when representing the directional attributes 'north', 'east', 'south' and 'west' using '1', '2', '3' and '4' respectively, the neighborhood relationship among the attributes is lost.

Nominal data

Numeric Conversion / Data encoding

- **Binary coding**: Depending on the number of representing bits used, the binary coding offers a rich sources to represent nominal attributes in terms of binary numbers.
 - *binary-coded decimal, n-ary grey codes, and one-hot encoding.*
 - Sophisticated coding schemes take into account the **probability distribution** of each attributes during conversion.
 - After conversion, the binary feature vectors can be compared based on various distance measures such as Hamming distance and Spearman distance.

Ordinal data

Examples

- In the patient queue management system, patients who are either in pain or in frail health are often given a higher **priority** than those without severe symptoms.
- In a typical survey, several levels of response preferences are often assigned to represent the **order** of preference in the observation.

Ordinal data

Numeric Conversion / Data encoding

- Similar to the nominal data, the percentage or the **frequency of occurrence can be quantified**.
- Unlike the nominal case, the ***mean, median and mode* values can be calculated** when the ranks are assigned a numerical value.
- However, care should be taken for using the *mean* value since there is **no knowledge regarding the distance among the ranks**.

Ordinal data

Numeric Conversion / Data encoding

- **Rank encoding with normalization:** The rating and ranking orders can be transformed into quantitative variables through normalization. Suppose the ranks are given by $r = 1, \dots, R$. Then, the ranks can be normalized into **standardized distance** values (d) which fall within $[0, 1]$ using

$$d = \frac{r - 1}{R - 1}. \quad (1)$$

- For comparing the **similarity** or dissimilarity between two ranks, several distances such as **Spearman distance**, **Hamming distance**, and **Chebyshev distance** can be used.

Ordinal data

Numeric Conversion / Data encoding

- The **Spearman distance** is proportional to the square of Euclidean distance between two rank vectors. For example, given that the preferences of person x and person y on fruits given by [1:apple, 2:orange, 3: kiwi] are respectively [2,3,1] and [3,2,1]. Then, the Spearman distance of preference between x and y is proportional to

$$\begin{aligned}d(x, y) &= (x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2 \\ &= (2 - 3)^2 + (3 - 2)^2 + (1 - 1)^2 = 2.\end{aligned}\tag{2}$$

- The **Hamming distance** between two strings of ranks of equal length is the number of positions at which the corresponding ranks are different. For example, the Hamming distance between [2,3,1] and [3,2,1] is 2 since the first two ranks values are different.
- The **Chebyshev distance** is also called *maximum value distance* which measures the maximum absolute difference between two ordinal vectors.
- Other distances for ordinal variables include Kendall distance, Cayley distance, and Ulam distance.

Interval and ratio data

The ratio scale has an absolute zero-point. For example, age, height and weight never fall below zero. For the interval scale, the zero-point is arbitrary where negative values is permitted.

Example

- The temperature degree in Celsius is considered an interval attribute, since the “zero-degree” does not correspond to “no temperature” where we cannot say that 60 degrees Celsius is twice as hot as 30 degrees Celsius.
- However, for the Kelvin scale, the zero point represents the complete absence of molecular motion where we can say that 60 Kelvins is twice as hot as 30 Kelvins. Hence, the Kelvin scale is a ratio attribute.
- Although a distance measure between two points can be made for both ratio and interval attributes, scaling of these two attributes produces different interpretations.
- Hence, for these two reasons (absolute zero value, and the interpretation of scales), special attention needs to be paid for the effect of scaling during data normalization.

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Distance metrics

Distance metrics

Definition

A *metric* or a *distance function* refers to a function that defines a distance between each pair of elements of a set. Mathematically, the distance mapping function can be written as

$$d(x, y) : \mathcal{X} \times \mathcal{X} \mapsto [0, \infty), \quad (3)$$

where $[0, \infty)$ is the set of nonnegative real numbers. The following conditions are satisfied

$$\text{Non-negativity axiom:} \quad d(x, y) \geq 0, \quad (4)$$

$$\text{Identity of indiscernibles:} \quad d(x, y) = 0 \Leftrightarrow x_1 = x_2, \quad (5)$$

$$\text{Symmetry:} \quad d(x, y) = d(y, x), \quad (6)$$

$$\text{Triangle inequality:} \quad d(x, z) \leq d(x, y) + d(y, z), \quad (7)$$

for all $x, y, z \in \mathcal{X}$. A set with a metric is called a metric space.

Distance metrics

Fig. 4 illustrates the three distances between each pair of points x , y , z .

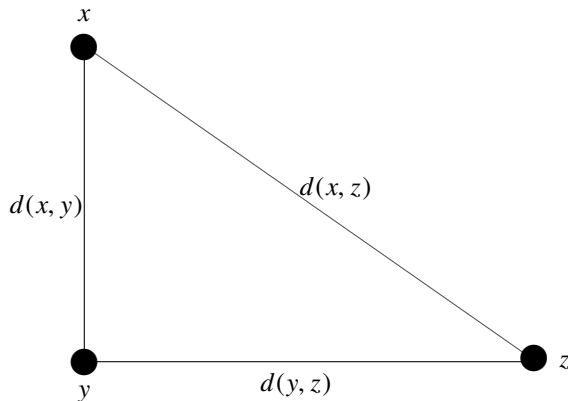


Figure 4: Three points form a triangle: $d(x, z) \leq d(x, y) + d(y, z)$.

Distance metrics

- The **Hamming distance** is a distance metric since it fulfills all the four metric conditions.
- Common examples of distance metrics include the **Euclidean metric** and the Manhattan or **taxicab metric**.
 - The Euclidean metric is also called the 2-norm or L_2 -norm distance. It is similar to the geometrical distance between two points in space.
 - The taxicab metric is also call the 1-norm or L_1 -norm distance.
- A generalization of both the Euclidean distance and the taxicab distance is called the **Minkowski distance** or p -norm metric.

Distance metrics

1-norm distance: $\sum_{i=1}^d |x_i - y_i|$

2-norm distance: $\left(\sum_{i=1}^d |x_i - y_i|^2 \right)^{1/2}$

p -norm distance: $\left(\sum_{i=1}^d |x_i - y_i|^p \right)^{1/p}$

infinity-norm distance: $\lim_{p \rightarrow \infty} \left(\sum_{i=1}^d |x_i - y_i|^p \right)^{1/p}$
 $= \max(|x_1 - y_1|, |x_2 - y_2|, \dots, |x_d - y_d|).$

The Minkowski distance is a metric for $p \geq 1$ since it satisfies all the four conditions listed in above. However, the triangle inequality is not satisfied for $p < 1$. Hence, the Minkowski distance is not a metric for $p < 1$.

Distance metrics

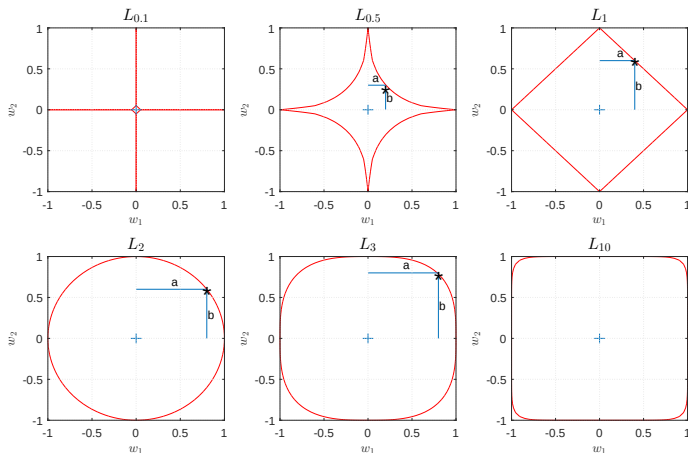


Figure 5: Minkowski distance (except for $L_{0.1}$ and $L_{0.5}$ which violate the triangle inequality, all other distances are distance metrics) on a two-dimensional space $\mathbf{w}$. The red solid contour indicates one unit distance from the origin. For example, in the L_1 plot, the distance between '+' and '*' is $d(+, *) = a + b = 1$. Other examples include $L_{0.5}$:

$d(+, *) = (a^{0.5} + b^{0.5})^{1/0.5} = 1$; L_2 : $d(+, *) = \sqrt{a^2 + b^2} = 1$; and L_3 : $d(+, *) = (a^3 + b^3)^{1/3} = 1$.

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Preprocessing Methods

Preprocessing

The collected **data can come in any of the above forms** namely, nominal, ordinal, interval and ratio types.

Moreover, **multiple data sources** can be collected in parallel and over a period of time. In other words, the data can come in large spatial dimensions and over time.

In order to achieve **consistency in data representation**, the following techniques are often adopted for data preprocessing.

Cleaning or cleansing

Data cleaning or cleansing is the process of detecting, correcting and removal of corrupted, incomplete, erroneous or inaccurate data samples from a collection.

- **Completeness**: This is the degree to which all required measurements are available. For some applications, the experiments can be repeated to collect more measurements. For others, those data with incomplete measurement are frequently removed.
- **Consistency**: This is the degree to which the measurements are reproducible under similar conditions. For stationary data, the measurements should not drift. For non-stationary data, the drift can be modelled.

Cleaning or cleansing

Data cleaning or cleansing is the process of detecting, correcting and removal of corrupted, incomplete, erroneous or inaccurate data samples from a collection.

- **Uniformity**: This is the degree to which the measurements are made using the same unit of scale and representation. For example, in one system the weight measurement could be in kilogram but in another system the weight measurement could be in pounds. If these data are to be put together, they have to be represented under the same scale.
- **Validity**: This is the degree to which the measurements conforms to well-established constraints. For example, for data represented in percentage, the total sum of such measurements must be equal to 100 percent.

Alignment

In [image processing](#), a frequent preprocessing step is to extract relevant portion for subsequent processing steps.

For example, in face recognition, the [face image](#) excluding the hairs and wearing accessories is often extracted from the [background](#) image for comparison. Here, the [alignment](#) of face is often obtained based on the eyes, nose and mouth locations where only regions around these regions are used for comparison.

Normalization

- **Min-max scaling:** When the **bounds or range of each independent dimension of data is known**, a common normalization technique is *min-max*. Alternatively, an estimation of the minimum and maximum values for a set of measurements can be made if the bounds are unknown. Given a set of raw measurements $\{x_i^{raw}\} \in [x^{min}, x^{max}]$, $i = 1, 2, \dots, M$. Then the normalized measurements can be computed using

$$x_i = \frac{x_i^{raw} - x^{min}}{x^{max} - x^{min}}, \quad i = 1, 2, \dots, M. \quad (8)$$

Normalization

- **Standardization**: When the population of measurements of each **independent dimension of data is normally distributed** where the parameters are known, the *standard score* or **z-score standardization** is a popular choice.

$$x_i = \frac{x_i^{raw} - E[X]}{\sigma(X)}, \quad i = 1, 2, \dots, M. \quad (9)$$

The exact values of $E[X]$ and $\sigma(X)$ may not be known in practice. Here, an estimation of the parameters can be made using a set of collected measurements:

$$\text{Sample mean : } \hat{\mu} = \frac{1}{M} \sum_{i=1}^M x_i, \quad (10)$$

$$\text{Sample variance : } \hat{\sigma} = \frac{1}{M} \sum_{i=1}^M (x_i - \hat{\mu})^2. \quad (11)$$

This normalization process is also called **standardization** in statistics.

Normalization

- **Median Absolute Deviation:** Instead of using the mean value as the reference point in standard deviation, the *median value* is adopted as reference point in the *median absolute deviation*:

$$\text{MAD} = \text{median}(|x_i - \text{median}(X)|), \quad (12)$$

where the normalization takes the form of

$$x_i = \frac{x^{raw} - \text{median}(X)}{\text{MAD}}. \quad (13)$$

Other Transformations

Apart from the above normalization schemes, other types of **data transformation** may be adopted to translate or stretch the data.

A simple transformation could be by means of translating the data towards a certain desired range of operation.

Operations such as multiplying, adding or subtracting a certain value **uniformly** to or from the measurement to get the data falling within a certain range provides a viable option to transform the data linearly.

Other Transformations

For measurements which spreads in exponential scale, taking the *logarithm* on such measurements (i.e., $\log(x_i)$) would be helpful to **scale down** the masking impact of large data values over small data values. Conversely, an exponential function (e.g., $\exp(x_i)$) can **stretch the data** when the measurements display a combination of flat and steep changing regions.

Some other nonlinear stretching include *sigmoid* and hyperbolic tanh functions:

$$x_i = \frac{1}{1 + e^{-h(x_i^{raw})}}, \quad (14)$$

$$x_i = \tanh(h(x_i^{raw})), \quad (15)$$

where the function $h(\cdot)$ can take any of the above normalization forms (such as (9)).

A Note on Normalization and Data Preprocessing for Training and Validation/Test

Performing global preprocessing (like calculating the global mean and variance for normalization) across the entire dataset before splitting leads to **data leakage** (or train-test contamination). Normalization parameters must be computed only on the training set and then applied to the validation/test sets.

Summary

- Data: a collection of **discrete or continuous values** that **convey information** in our context.
- Data types:
 - **Nominal**: categorical or qualitative
 - **Ordinal**: have an explicit order, but the exact differences between the ranks are unknown
 - **Interval and ratio**: represented by numerics
- Distance metrics: L_1 -norm, L_2 -norm, L_p -norm and etc
- Data normalization: **min-max scaling**, **z-score standardization**, etc

Thank you!